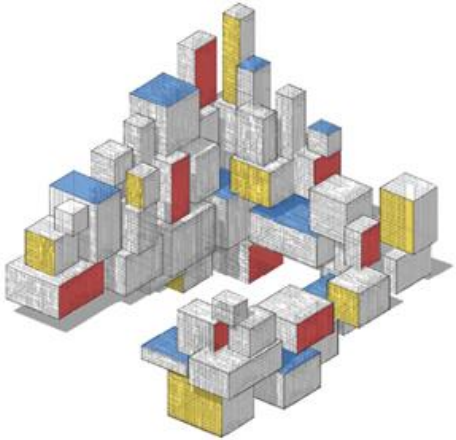
Chapter 3
Statistical Learning



Instructor: Dr. Heyfa Ammar

$$MSE = \frac{1}{n} \sum_i (y - y_pred)^2 = np.mean(y - y_pred)^2$$

$$\underset{\text{Mean Absolute Error}}{MAE} = \frac{1}{n} \sum_i |y - y_pred|$$

$$\underset{\text{Root square}}{\underset{\sim}{R}MSE} = \sqrt{MSE}$$

OVERVIEW

Purpose

We introduce common concepts and themes in statistical learning.

S

Supervised Learning

Learn a functional relationship between features and a labeled response, guided by a training set.

U

Unsupervised Learning

Learn the structure of the data's distribution itself, with no response variable.

A

Assessing Performance

Quantify how well a learner will predict on new, unseen data.

What is Statistical Learning?

A main challenge in data science is the mathematical analysis of the data.

When the goal is to interpret the model and quantify the uncertainty in the data, this analysis is usually referred to as statistical learning.

There are two major goals for modeling data:

- To accurately predict some future quantity of interest, given some observed data.
- To discover unusual or interesting patterns in the data.

Two Goals for a Model

Both models can predict a house price — but only one tells you why, and how sure it is.



Black-Box Prediction

"$450,000."

Accurate, but can't say why — and no sense of how confident the number is.



Statistical Learning

"+100 sq ft → +$15,000, ±$3,000."

Explains the relationship AND quantifies how confident we are in it.

"When the goal is to interpret the model and quantify the uncertainty in the data, this analysis is usually called statistical learning."

Example: Predicting House Prices

$$\text{Price} = \beta_0 + \beta_1(\text{sq ft}) + \beta_2(\text{bedrooms}) + \beta_3(\text{neighborhood})$$

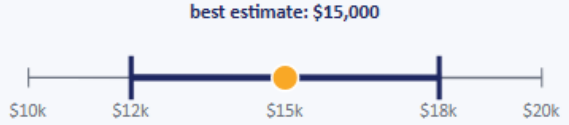
INTERPRETATION

Each extra 100 sq ft is associated with about a $15,000 increase in price, holding bedrooms and neighborhood fixed.

UNCERTAINTY

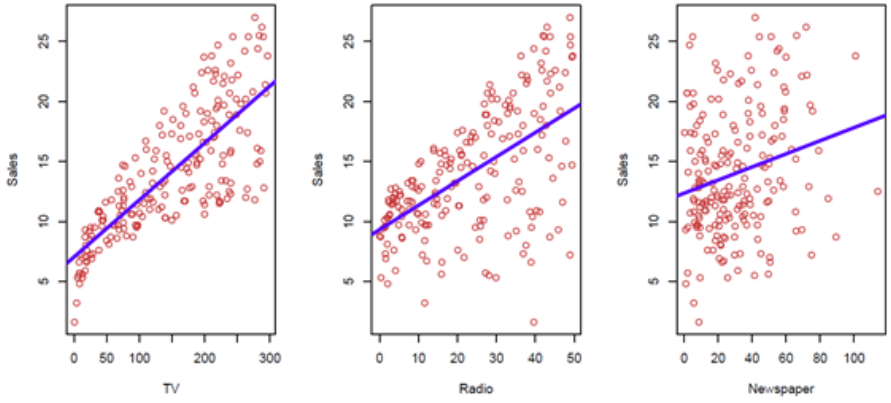
We're 95% confident the true effect is between $12,000 and $18,000 — not just a single guess.

EFFECT OF +100 SQ FT ON PRICE



The dot is our best guess. The bar shows the range we're confident the true effect falls within.

What is Statistical Learning?



Shown are `Sales` vs `TV`, `Radio` and `Newspaper`, with a blue linear-regression line fit separately to each.
Can we predict `Sales` using these three?
Perhaps we can do better using a model

$$\texttt{Sales} \approx f(\texttt{TV}, \texttt{Radio}, \texttt{Newspaper})$$

What is Statistical Learning?

Here `Sales` is a *response* or *target* that we wish to predict. We generically refer to the response as Y.
`TV` is a *feature*, or *input*, or *predictor*; we name it X_1.
Likewise name `Radio` as X_2, and so on.
We can refer to the *input vector* collectively as

$$X = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}$$

Now we write our model as

$$Y = f(X) + \epsilon$$

where ϵ captures measurement errors and other discrepancies.

Statistical Learning

- With a good f we can make predictions of Y at new points $X = x$.
- We can understand which components of $X = (X_1, X_2, \ldots, X_p)$ are important in explaining Y, and which are irrelevant. e.g. `Seniority` and `Years of Education` have a big impact on `Income`, but `Marital Status` typically does not.
- Depending on the complexity of f, we may be able to understand how each component X_j of X affects Y.

Regression and Classification

In regression problems, the response variable y can take any real value.

In contrast, when y can only lie in a finite set, say $y \in \{0, \ldots, c-1\}$, then predicting y is conceptually the same as classifying the input $\boldsymbol{x}$ into one of c categories, and so prediction becomes a classification problem.

Loss Function

We can measure the accuracy of a prediction $\widehat{y}$ with respect to a given response y by using some loss function $\text{Loss}(y, \widehat{y})$.

In a regression setting the usual choice is the squared-error loss $(y - \widehat{y})^2$.

In the case of classification, the zero–one (also written 0–1) loss function $\text{Loss}(y, \widehat{y}) = \mathbb{I}\{y \neq \widehat{y}\}$ is often used, which incurs a loss of 1 whenever the predicted class $\widehat{y}$ is not equal to the class y.

Later on we will encounter various other useful loss functions, such as the cross-entropy and hinge loss functions.

Error

The word error is often used as a measure of distance between a "true" object y and some approximation $\widehat{y}$ thereof.

If y is real-valued, the absolute error $|y - \widehat{y}|$ and the squared error $(y - \widehat{y})^2$ are both well-established error concepts

For vectors, we have instead the norm $\|\boldsymbol{y} - \widehat{\boldsymbol{y}}\|$ and squared norm $\|\boldsymbol{y} - \widehat{\boldsymbol{y}}\|^2$ for.

The squared error $(y - \widehat{y})^2$ is just one example of a loss function.

Risk

It is unlikely that any mathematical function g will be able to make accurate predictions for all possible pairs $(\boldsymbol{x}, y)$.

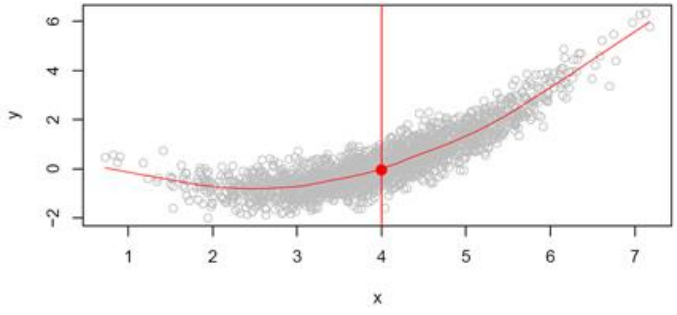
Even with the same input $\boldsymbol{x}$, the output y may be different, depending on chance circumstances or randomness.

For this reason, we adopt a probabilistic approach and assume that each pair $(\boldsymbol{x}, y)$ is the outcome of a random pair $(\boldsymbol{X}, Y)$ that has some joint probability density $f(\boldsymbol{x}, y)$.

We then assess the predictive performance via the expected loss, usually called the risk, for g:

$$\ell(g) = \mathbb{E}\,\mathrm{Loss}(Y, g(\boldsymbol{X})). \tag{1}$$

Statistical Learning



Is there an ideal $f(X)$? In particular, what is a good value for $f(X)$ at any selected value of X, say $X = 4$? There can be many Y values at $X = 4$. A good value is

$$f(4) = E(Y|X = 4)$$

$E(Y|X = 4)$ means *expected value* (average) of Y given $X = 4$.

This ideal $f(x) = E(Y|X = x)$ is called the *regression function*.

Statistical Learning

The regression function $f(x)$

- Is also defined for vector X; e.g.
  $f(x) = f(x_1, x_2, x_3) = E(Y|X_1 = x_1, X_2 = x_2, X_3 = x_3)$
  True value & depends on x_1, x_2, x_3
- Is the *ideal* or *optimal* predictor of Y with regard to mean-squared prediction error: $f(x) = E(Y|X = x)$ is the function that minimizes $E[(Y - g(X))^2|X = x]$ over all functions g at all points $X = x$.
- $\epsilon = Y - f(x)$ is the *irreducible* error — i.e. even if we knew $f(x)$, we would still make errors in prediction, since at each $X = x$ there is typically a distribution of possible Y values.
- For any estimate $\hat{f}(x)$ of $f(x)$, we have

$$E[(Y - \hat{f}(X))^2|X = x] = \underbrace{[f(x) - \hat{f}(x)]^2}_{\text{Reducible}} + \underbrace{\text{Var}(\epsilon)}_{\text{Irreducible}}$$

approximation of a true model (loss function) some error

set of all outputs of the variables; set of features

[we can reduce the error by finding a better model] so try to make the error less

Optimal Prediction Function

we have multiple models, for each iteration I'll find the loss then choose the best function overall with the least error / minimum expected loss / minimum average

Given the distribution of $(\boldsymbol{X}, Y)$ and any loss function, the goal is to find the optimal prediction function that yields the smallest risk:

$$g^* = \underset{g}{\operatorname{argmin}} \underbrace{\mathbb{E}\,\mathrm{Loss}(Y, g(\boldsymbol{X}))}_{\ell(g)}.$$

In the classification case with zero–one loss function the risk is equal to the probability of incorrect classification: $\ell(g) = \mathbb{P}[Y \neq g(\boldsymbol{X})]$.

true classification is different to the predicted classification

In this context, the prediction function g is called a classifier.

Training set

Since, the optimal prediction function g^* depends on the typically unknown joint distribution of $(\boldsymbol{X}, Y)$, it is not available in practice.

True x, True y, but in practice we don't have these true values

Instead, all that we have available is a finite number of (usually) independent realizations from the joint density $f(\boldsymbol{x}, y)$.

like 1000 patients (sample)

We denote this sample by $\mathcal{T} := \{(\boldsymbol{X}_1, Y_1), \ldots, (\boldsymbol{X}_n, Y_n)\}$ and call it the training set ($\mathcal{T}$ is a mnemonic for training) with n examples.

training set; composed of x (predictors) and y (output)

It will be important to distinguish between a random training set $\mathcal{T}$ and its (deterministic) outcome $\tau := \{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n)\}$.

Supervised Learning

In supervised learning we try to learn the functional relationship g^* between the feature vector (explanatory variable) $\boldsymbol{x}$ and response y in the presence of a teacher who provides a training set $\mathcal{T}$ with n examples. A teacher giving the exercise and the solution (x and Y are known)

The output is known like which patients have heart diseases and which dont

Denote by $g_{\mathcal{T}}$ the best (by some criterion) approximation for g^* that we can construct from $\mathcal{T}$.

Note that this so-called learner $g_{\mathcal{T}}$ is a random function. A particular outcome is denoted by g_τ.

The main supervised learning methodologies are regression and classification.

More advanced supervised learning techniques, including reproducing kernel Hilbert spaces, tree methods, and deep learning.

Unsupervised Learning

In unsupervised learning we make no distinction between response and explanatory variables, and the objective is simply to learn the structure of the unknown distribution of the data.

Objective is learn the structure of the data

In other words, we need to learn $f(\boldsymbol{x})$.

In this case the guess $g(\boldsymbol{x})$ is an approximation of $f(\boldsymbol{x})$ and the risk is of the form

$$\ell(g) = \mathbb{E}\,\text{Loss}(f(\boldsymbol{X}), g(\boldsymbol{X})).$$

The main methodologies for unsupervised learning include clustering, principal component analysis, and kernel density estimation.

The output isn't known

The answer isn't given but the model learns through something like we give it clusters of different picture like one has groups of laptop pics and it learn by itself through some data given

Training Loss

The loss while training

Given an arbitrary prediction function g, it is typically not possible to compute its risk $\ell(g) = \mathbb{E}\,\mathrm{Loss}(Y, g(\boldsymbol{X}))$.

average loss on the whole population which is hard to know

However, using the training sample $\mathcal{T}$, we can approximate $\ell(g)$ via the empirical (sample average) risk

$$\ell_{\mathcal{T}}(g) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{Loss}(Y_i, g(\boldsymbol{X}_i)), \tag{2}$$

sample size

which we call the training loss.

The training loss is thus an unbiased estimator of the risk (the expected loss) for a prediction function g, based on the training data.

Error / Loss / Risk

Concept	What it is	Aggregation	Scope / based on	Example
Error	Distance between a true value and its approximation	none	single point, general/informal	$\lvert y - \hat{y} \rvert$
Loss Evaluating the model based on one row	A specific function quantifying error for one prediction	none (pointwise)	single observation	$(y - \hat{y})^2, \mathbb{I}\{y \neq \hat{y}\}$
Training/test loss taking a part of the data set (multiple rows) then finding the average	Sample average of the loss	sample average	a finite, known/observed sample	$\ell_\tau(g) = \frac{1}{n}\sum_{i=1}^{n} \text{Loss}(y_i, g(x_i))$
Risk The true distribution is usually unknown	Expected loss over the true data distribution	expectation (population average)	full, usually unknown distribution $f(x, y)$	$\ell(g) = \mathbb{E}\,\text{Loss}(Y, g(X))$

Learner

To approximate the optimal prediction function g^* (the minimizer of the risk $\ell(g)$) we

1. first select a suitable collection of approximating functions $\mathcal{G}$ and then
2. take our learner to be the function in $\mathcal{G}$ that minimizes the training loss; that is,

$$g_{\mathcal{T}}^{\mathcal{G}} = \underset{g \in \mathcal{G}}{\operatorname{argmin}}\, \ell_{\mathcal{T}}(g). \tag{3}$$

For example, the simplest and most useful $\mathcal{G}$ is the set of linear functions of $\boldsymbol{x}$; that is, the set of all functions $g : \boldsymbol{x} \mapsto \boldsymbol{\beta}^\top \boldsymbol{x}$ for some real-valued vector $\boldsymbol{\beta}$.

Overfitting

Note that minimizing the training loss over *all* possible functions g (rather than over all $g \in \mathcal{G}$) does not lead to a meaningful optimization problem, as any function g for which $g(\boldsymbol{X}_i) = Y_i$ for all i gives minimal training loss.

In particular, for a squared-error loss, the training loss will be 0. Unfortunately, such functions have a poor ability to predict new (that is, independent from $\mathcal{T}$) pairs of data. This poor generalization performance is called overfitting.

> By choosing g a function that predicts the training data exactly (and is, for example, 0 otherwise), the squared-error training loss is zero. Minimizing the training loss is not the ultimate goal!

Generalization Risk

The prediction accuracy of new pairs of data is measured by the generalization risk of the learner. For a fixed training set τ it is defined as

$$\ell(g_\tau^{\mathcal{G}}) = \mathbb{E}\,\mathrm{Loss}(Y, g_\tau^{\mathcal{G}}(\boldsymbol{X})), \tag{4}$$

where $(\boldsymbol{X}, Y)$ is distributed according to $f(\boldsymbol{x}, y)$.

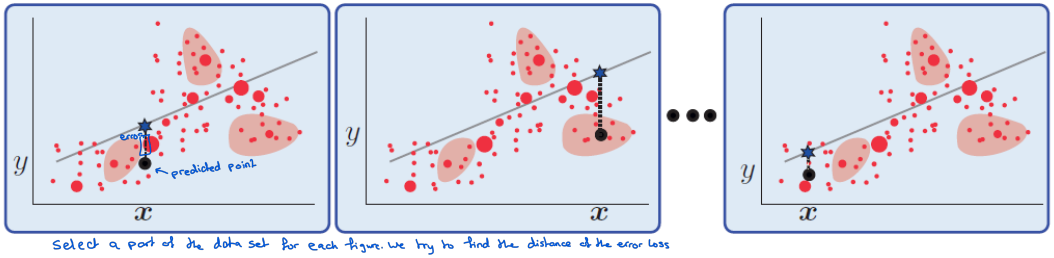


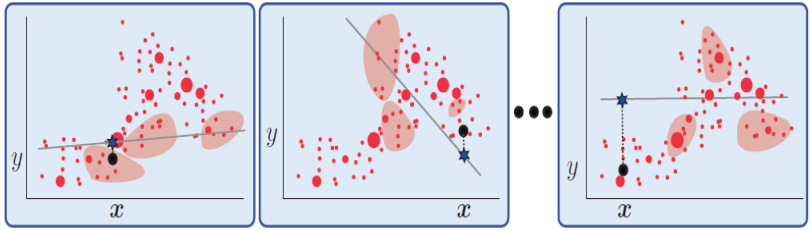
Figure: The generalization risk for a fixed training set is the weighted-average loss over all possible pairs $(\boldsymbol{x}, y)$.

Expected Generalization Risk

For a random training set $\mathcal{T}$, the generalization risk is thus a random variable that depends on $\mathcal{T}$ (and $\mathcal{G}$). If we average the generalization risk over all possible instances of $\mathcal{T}$, we obtain the expected generalization risk:

$$\mathbb{E}\,\ell(g_{\mathcal{T}}^{\mathcal{G}}) = \mathbb{E}\,\mathrm{Loss}(Y, g_{\mathcal{T}}^{\mathcal{G}}(\boldsymbol{X})), \tag{5}$$

where $(\boldsymbol{X}, Y)$ in the expectation above is independent of $\mathcal{T}$.



Some data but different training sets to find the best model using different iterations

Figure: The expected generalization risk is the weighted-average loss over all possible pairs $(\boldsymbol{x}, y)$ and over all training sets.

To compute this exactly, you'd need $f(x, y)$ itself — which is precisely the thing statistical learning is trying to learn *about* in the first place. It's circular: if you already knew the full distribution, you wouldn't need to learn a model from data at all.

This is exactly why the training/test loss exists — it's a practical workaround. Instead of averaging over the *true* distribution (impossible), you average over a *sample* drawn from that distribution (possible):

By the law of large numbers, as your sample size n grows, this sample average gets closer and closer to the true risk. That's the sense in which it's an "unbiased estimator" — on average across many possible samples, it hits the right value, and it converges to the true risk as $n \to \infty$.

Test Loss

The loss calculated on a test sample (selected randomly)

For any outcome τ of the training data, we can estimate the generalization risk without bias by taking the sample average

$$\ell_{\mathcal{T}'}(g_\tau^{\mathcal{G}}) := \frac{1}{n'} \sum_{i=1}^{n'} \text{Loss}(Y_i', g_\tau^{\mathcal{G}}(\boldsymbol{X}_i')), \tag{6}$$

where $\{(\boldsymbol{X}_1', Y_1'), \ldots, (\boldsymbol{X}_{n'}', Y_{n'}')\} =: \mathcal{T}'$ is a so-called test sample. The test sample is completely separate from $\mathcal{T}$, but is drawn in the same way as $\mathcal{T}$; that is, via independent draws from $f(\boldsymbol{x}, y)$, for some sample size n'. We call the estimator (6) the test loss.

For a random training set $\mathcal{T}$ we can define $\ell_{\mathcal{T}'}(g_{\mathcal{T}}^{\mathcal{G}})$ similarly. It is then crucial to assume that $\mathcal{T}$ is independent of $\mathcal{T}'$.

Test and Validation Sets

To compare the predictive performance of different learners (as measured by the test loss), we can use the same fixed training set τ and test set τ' for all learners.

When there is an abundance of data, the "overall" data set is usually (randomly) divided into a training and test set. If the latter is used for model selection, a third set is needed for testing the performance of the selected model.

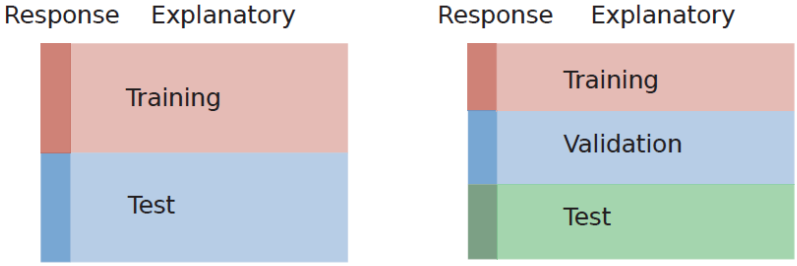


Figure: Statistical learning algorithms often require the data to be divided into various sets.

Notation

$\boldsymbol{x}$ and $\boldsymbol{X}$	Fixed and random explanatory (feature) vector.
y and Y	Fixed and random response.
$f(\boldsymbol{x}, y)$	Joint pdf of $\boldsymbol{X}$ and Y.
τ and $\mathcal{T}$	Fixed and random training data $\{(\boldsymbol{X}_i, Y_i)\}$.
$\mathbf{X}$	Matrix of explanatory variables, with rows $\{\boldsymbol{x}_i^\top\}$.
g	Prediction (guess) function.
$\text{Loss}(y, \widehat{y})$	Loss incurred when predicting response y with $\widehat{y}$.
$\ell(g)$	Risk for g, i.e., $\mathbb{E}\,\text{Loss}(Y, g(\boldsymbol{X}))$.
g^*	Optimal prediction function; i.e., $\text{argmin}_g\, \ell(g)$.
$\ell_\tau(g)$	Training loss for g, i.e., the sample average estimate of $\ell(g)$ based on τ.
$g_\tau^{\mathcal{G}}$	The learner: $\text{argmin}_{g \in \mathcal{G}}\, \ell_\tau(g)$.

Finding Good Learners

To find good learners, one must rely on knowledge from three important pillars of the mathematical sciences.

1. **Function approximation.** This is a natural way to represent a relationship between variables. We usually assume that this mathematical function is not completely known, but can be approximated well given enough computing power and data.
2. **Optimization.** Given a class of approximating functions, we wish to find the best possible function in that class. This requires some kind of efficient search or optimization procedure.
3. **Probability and Statistics.** In general, the data used to fit the model is viewed as a realization of a random process or numerical vector, whose probability law determines the accuracy with which we can predict future observations.

Example

Data: $(U_i, Y_i), i = 1, \ldots, 100$, where the $\{U_i\} \sim_{\text{iid}} \mathcal{U}(0, 1)$ and, given $U_i = u_i$, $Y_i \sim \mathcal{N}(10 - 140u_i + 400u_i^2 - 250u_i^3,\ 25)$.

